

ORF307: Optimization

Precept: #3

Agenda

- Flop count
- Least squares
- Positive (Semi-)Definite Matrices and Gram matrices
- Least squares interpretation
- More coding practice (see notebook in Companion).

Reminders and Announcements

- Always check Python notebooks in Companion (in directories `lectures/` and `precepts/`)
- Write down the deadline of the current HW:
- Other announcements on blackboard.

Flop count

Consider $A, B \in \mathbb{R}^{m \times n}$, $C \in \mathbb{R}^{n \times p}$, $x \in \mathbb{R}^n$, and $\alpha \in \mathbb{R}$

Matrix operation	Flop count
αA	
$A + B$	
$x^T x$	
Ax	
AC	
$A^T A$	

Least squares

Consider the optimization problem

$$\min_x \|Ax - b\|_2^2$$

where the matrix $A \in \mathbb{R}^{m \times n}$ is tall (which means we have many more rows than columns and the system is **overdetermined**).

NOTE: When a number m is much larger than another number n we can use the notation: $m \gg n$.

Derive the normal equations:

Positive (semi) definite matrices

Definition: A matrix $A \in \mathbb{R}^{n \times n}$ is called positive semidefinite matrix (PSD) iff

Definition: A matrix $A \in \mathbb{R}^{n \times n}$ is called positive definite matrix (PD) iff

Gram matrices:

The matrix $A^T A$ is called the **Gram matrix** of A .

Show that $A^T A$ has the following properties:

- $A^T A$ is symmetric matrix.

Proof:

- $A^T A \succ 0$ (i.e., $A^T A$ is Positive Definite (PD) matrix) if the columns of A are linearly independent.

Proof:

Least squares interpretation

Interpret the least squares problem and its solution using the following example:

$$\min \|Ax - b\|_2^2$$

where

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad b = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Least squares orthogonality principle:

Prove that the residual vector $r = Ax^* - b$ is orthogonal to any vector $Ax, \forall x \in \mathbb{R}^n$. In other words, we always have:

$$Ax \perp r, \forall x \in \mathbb{R}^n$$

Proof: